

① Electromagnetic Waves:

- These are transverse waves
- These can be polarized
- These are produced by accelerated charges
- These carry Energy and momentum ($\phi = \frac{h}{\lambda}$)

Intensity of EM wave

Energy flux $\rightarrow$ Poynting Vector

Physics Galaxy
Videos

$$\vec{S} = \frac{1}{\mu_0} (\vec{E} \times \vec{B})$$

W/m^2

- Energy density of EF & MF $\rightarrow u_B = \frac{B^2}{2\mu_0}$
 $\rightarrow u_E = \frac{1}{2}\epsilon_0 E^2$

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

Dir of prop of EM wave.

EM Spectrum

Radiowaves (few m to m.)

Microwaves (10^{-5}m to 10^{-4}m)

Infrared ($7800 - 10^5 \text{Å}$)

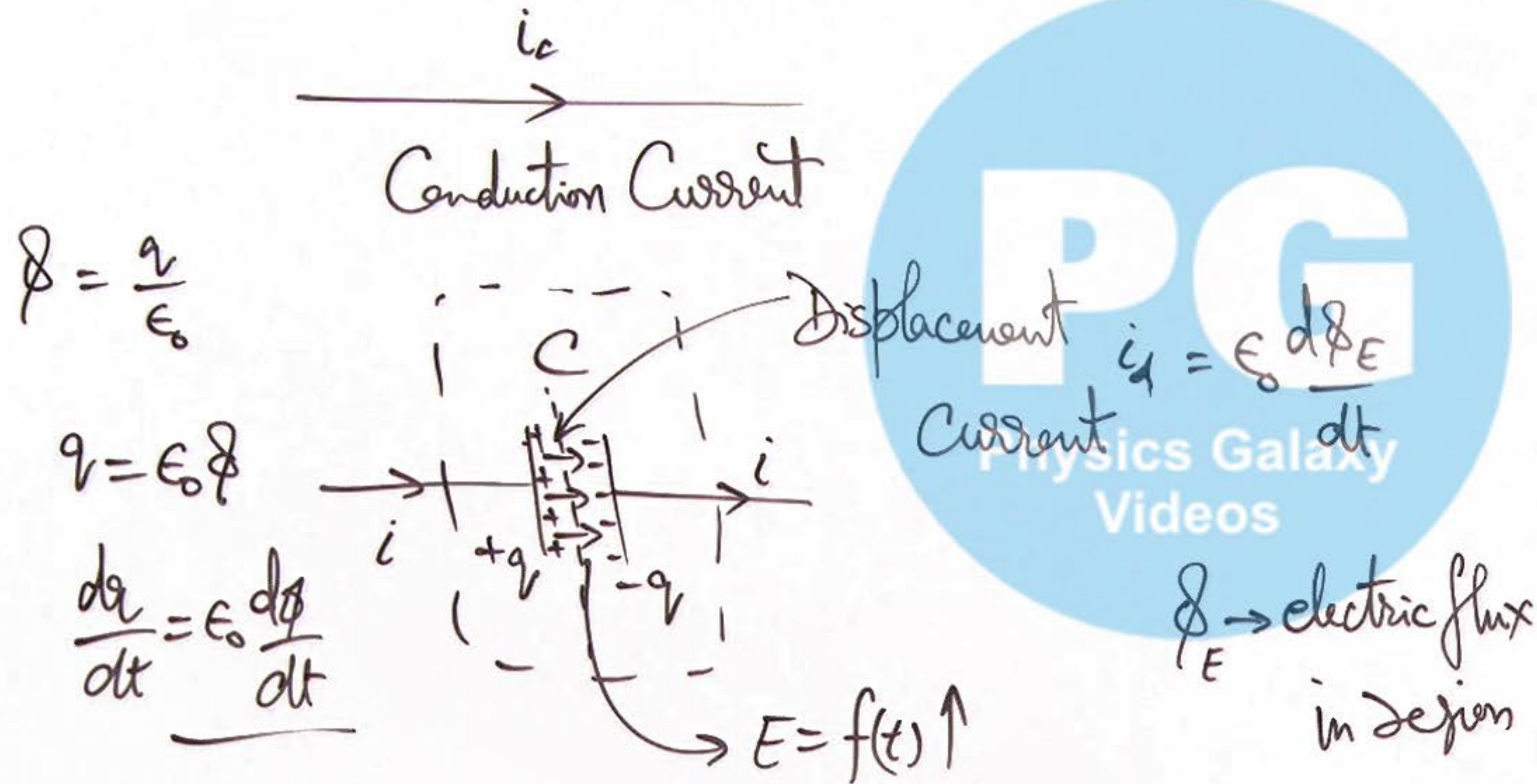
Visible ($4200 - 7800 \text{Å}$)

Ultraviolet ($100 - 4200 \text{Å}$)

X-Rays ($0.1 \text{Å} - 100 \text{Å}$)

γ -Rays ($< 0.1 \text{Å}$)

② Displacement Current :



③ Modification of Ampere's Law :

$$\oint_M \vec{B} \cdot d\vec{l} = \mu_0 i_{\text{enc.}}$$

Current enclosed by path M.

$$i_{\text{enc}} = i_c + i_d$$

$$\Rightarrow \oint_M \vec{B} \cdot d\vec{l} = \mu_0 i_c + \mu_0 \epsilon_0 \frac{d\phi_E}{dt}$$

④ Maxwell's Equations :

(i) Gauss's Law in Electrostatics

$$\Phi_E = \oint \vec{E} \cdot d\vec{S} = \frac{q_{\text{enc}}}{\epsilon_0} \quad - (1)$$

(ii) Gauss's Law in Magnetism

$$\Phi_M = \oint_M \vec{B} \cdot d\vec{S} = 0 \quad \left[\begin{array}{l} \text{Because of} \\ \text{closed loop lines} \end{array} \right] \quad - (2)$$

(iii) Faraday's law of EMI

$$\mathcal{E} = - \frac{d\Phi_B}{dt} = - \frac{d}{dt} \int \vec{B} \cdot d\vec{S}$$

(iv) Maxwell's Ampere law

$$\oint \vec{E} \cdot d\vec{l} = - \frac{d\Phi_B}{dt} \quad - (3)$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 i_c + \mu_0 \epsilon_0 \frac{d\Phi_E}{dt} \quad - (4)$$

⑤ Nature of EM Waves :

(i) Transverse Nature $\rightarrow$ $E_{ms} = E_0/\sqrt{2}$; $B_{ms} = \frac{B_0}{\sqrt{2}}$

$$\left. \begin{aligned} \vec{E} &= E_0 \sin(\omega t - kx) \hat{j} \\ \vec{B} &= B_0 \sin(\omega t - kx) \hat{k} \end{aligned} \right\} \text{Dir of EM wave}$$

$$\underline{\hat{C}} = (\vec{E} \times \vec{B}) = \underline{\hat{E} \times \hat{B}}$$

(ii) Relation in Amplitudes $\rightarrow$

$$E_0 = c B_0$$

(iii) Speed of EM waves $\rightarrow$

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} ; \quad C_m = \frac{1}{\sqrt{\mu_0 \mu_r \epsilon_0 \epsilon_r}} = \frac{c}{\sqrt{\mu_r \epsilon_r}}$$

(iv) Intensity of EM waves $\rightarrow$

$$I = \frac{1}{2} \epsilon_0 E_0^2 c = \epsilon_0 E_{ms}^2 c$$